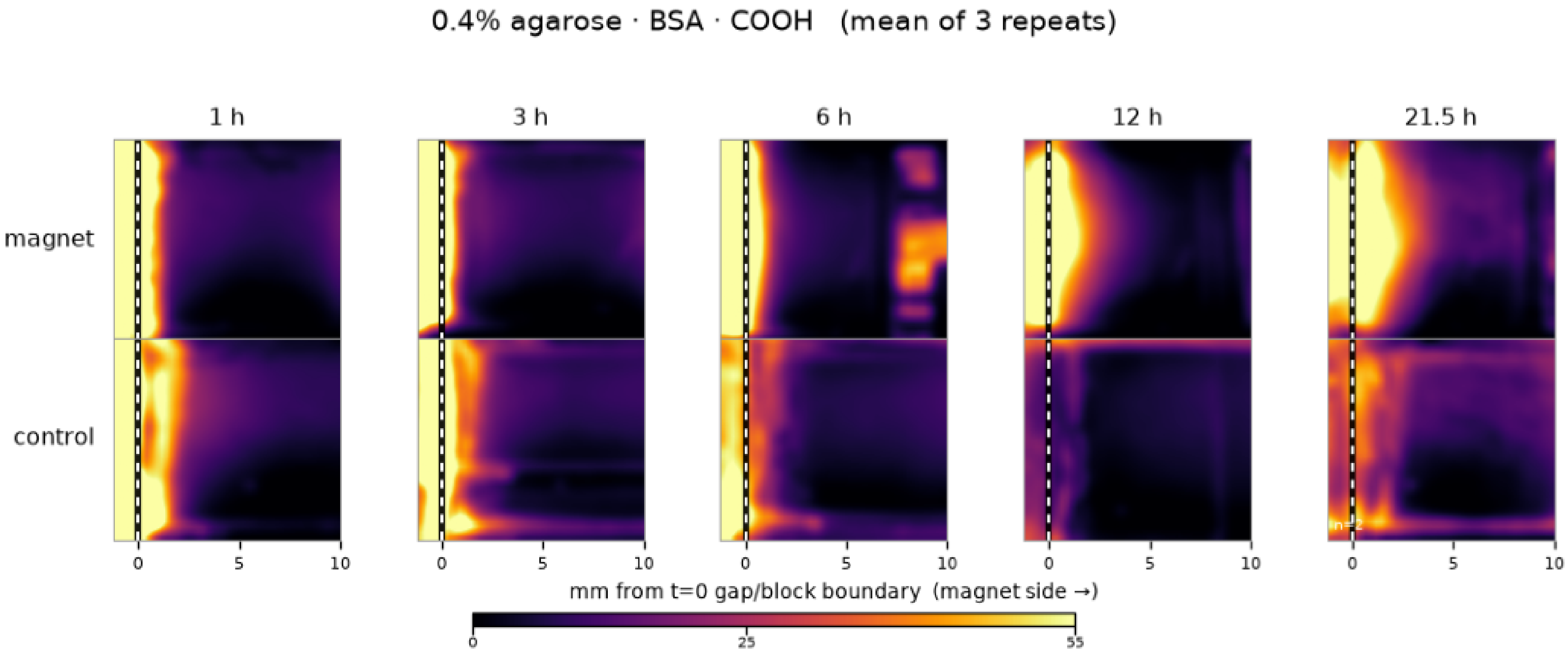
Magnetic Nanoparticle Transport in Agarose

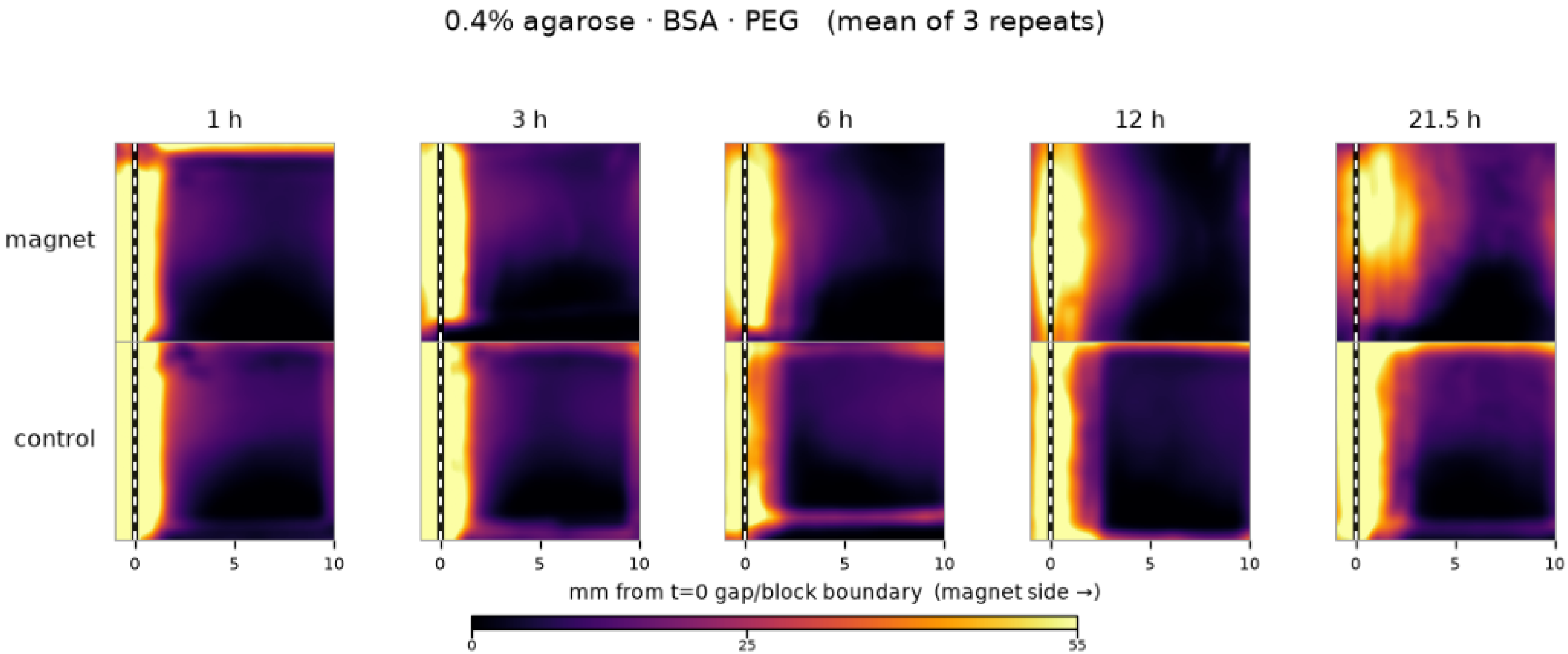
Block heatmaps — 21.5 h back-injection run, 27 Aug 2026

Each page shows one condition: the magnet arm (top row) against its no-magnet controls (bottom row), each cell the mean of 3 repeats, at 1, 3, 6, 12 and 21.5 hours. Every cell is the 10×10 mm agarose block seen from above — injection gap at the left, magnet side at the right. Color is NP darkening above each frame's own gel floor (camera exposure cancelled); the white dashed line marks the gap/block boundary measured at t = 0 and held fixed (agarose shrinkage is shown, not compensated). One shared color scale (0–55 L*) across all pages, so any two cells are directly comparable.

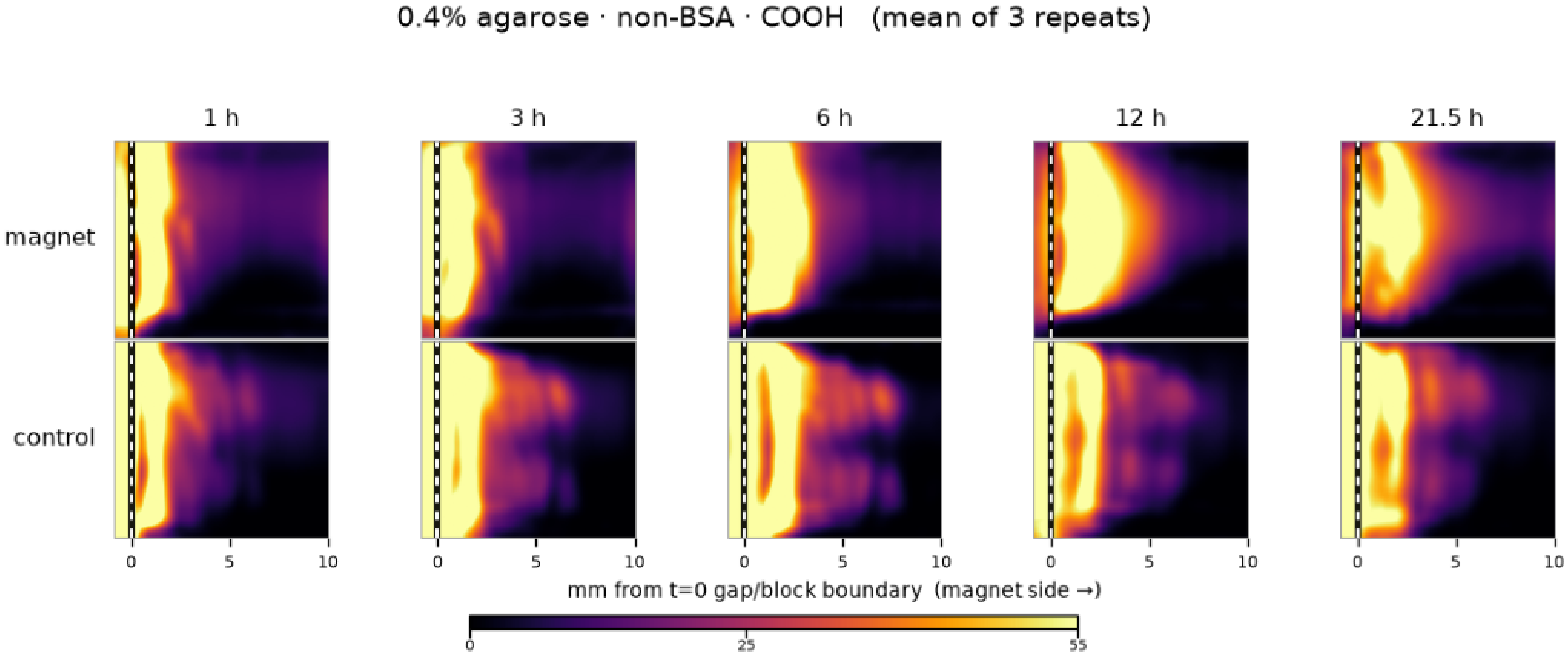
Frames are registered onto their series' t = 0 image and anchored on the case's rim edges (both sides) and the injection gap. In magnet rows, the magnet's shadow adds some signal toward the right edge (deepening with ambient light changes; the 12 h photos were taken at night) — compare against the control row beneath.

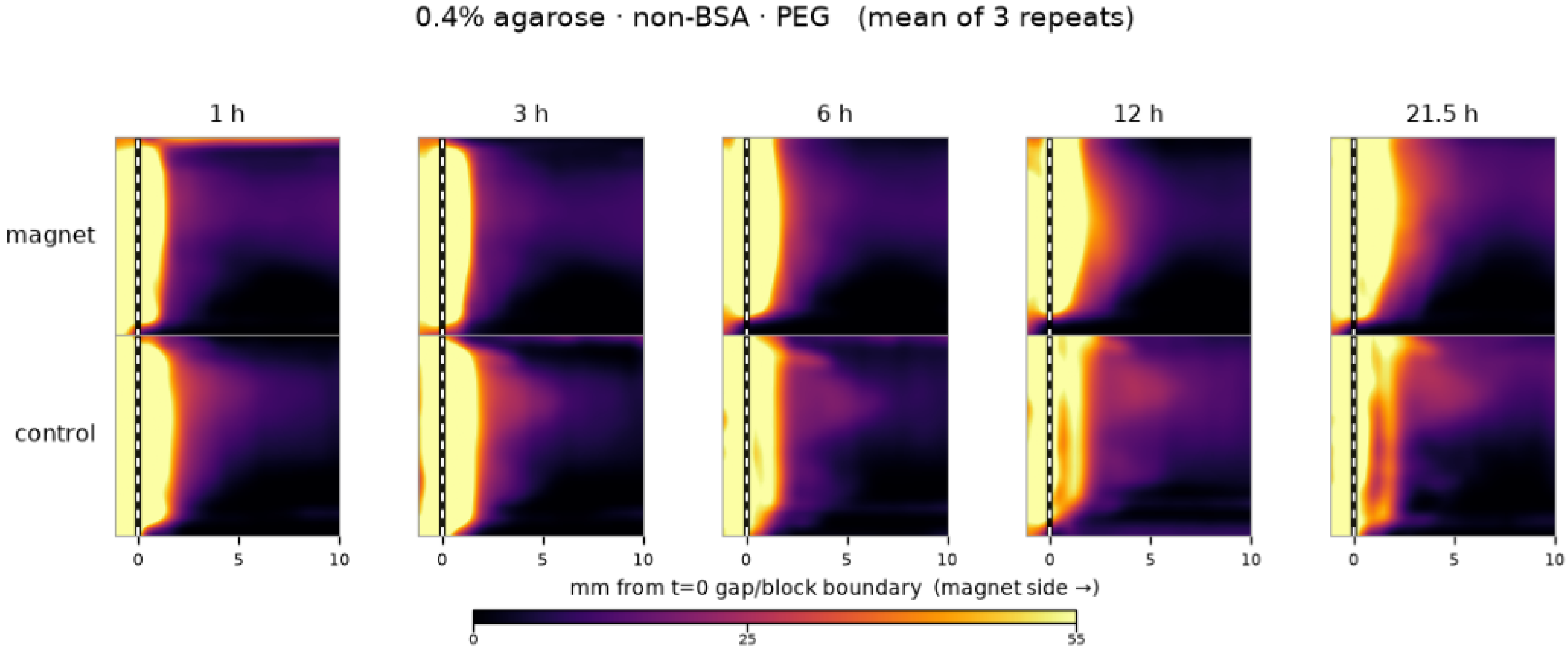
github.com/sthoa/nanoparticles

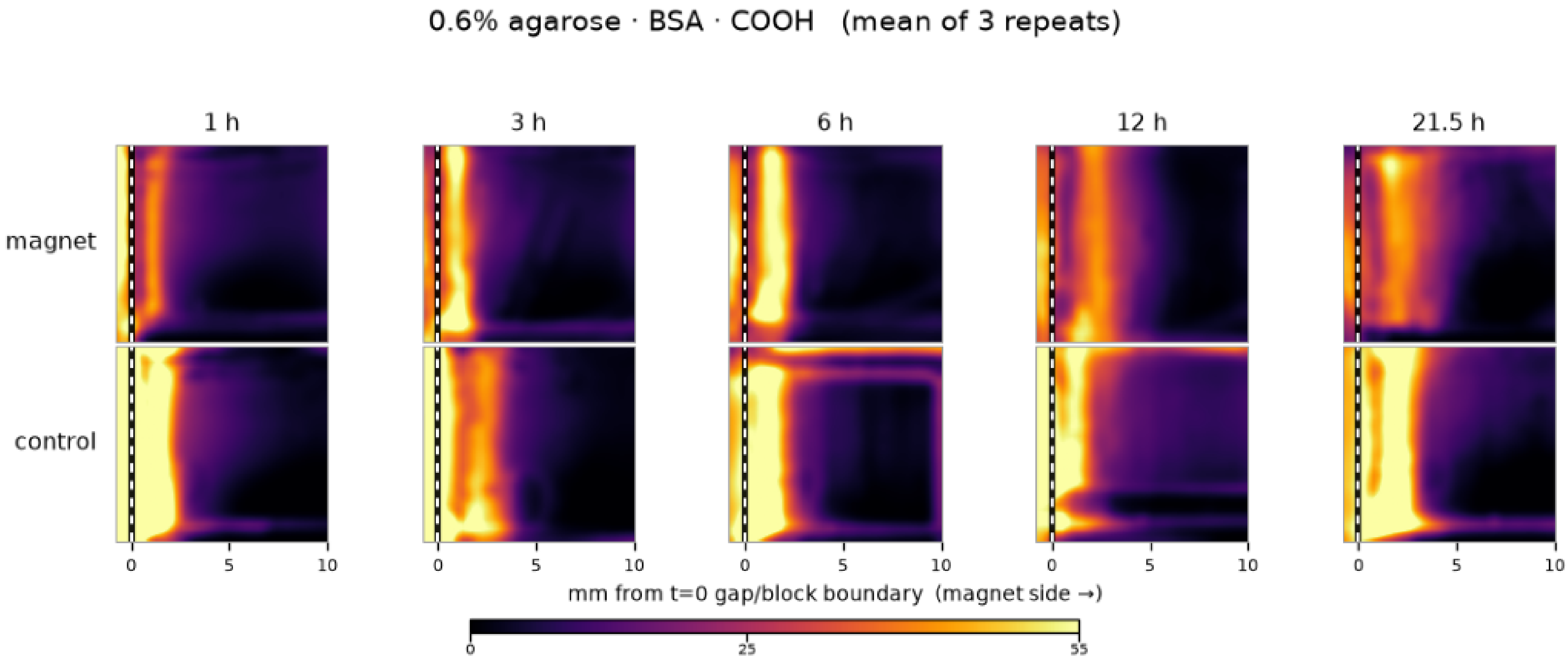

0.4% agarose · BSA · COOH (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
n=2
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55

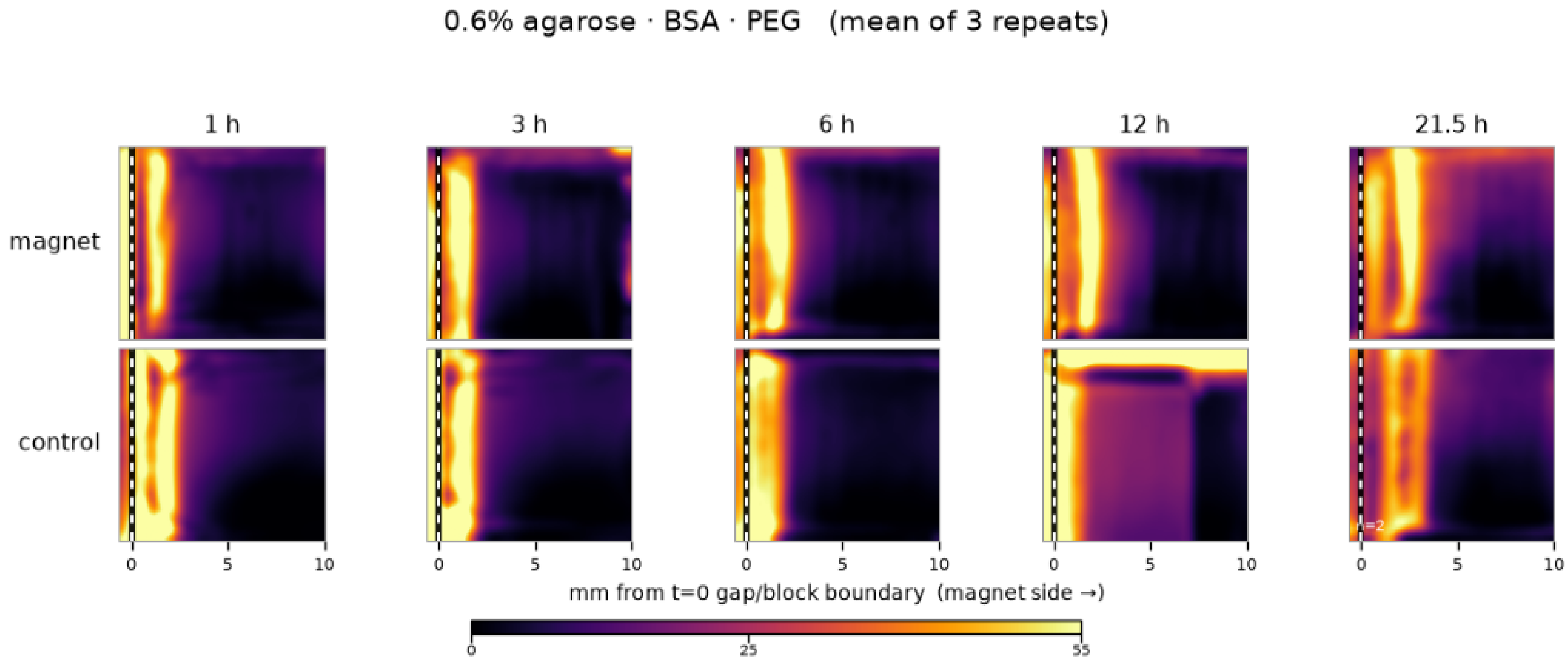


0.4% agarose · BSA · PEG (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55

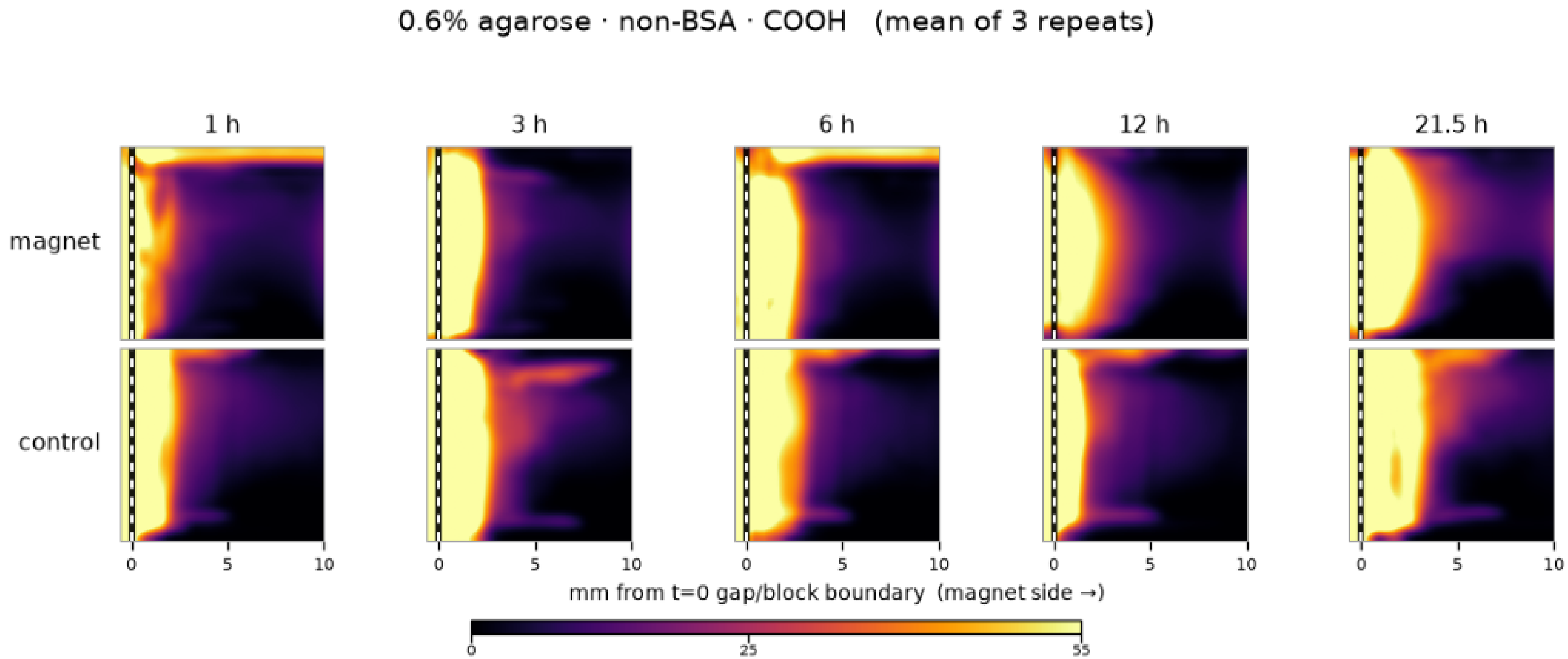

0.4% agarose · non-BSA · COOH (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55

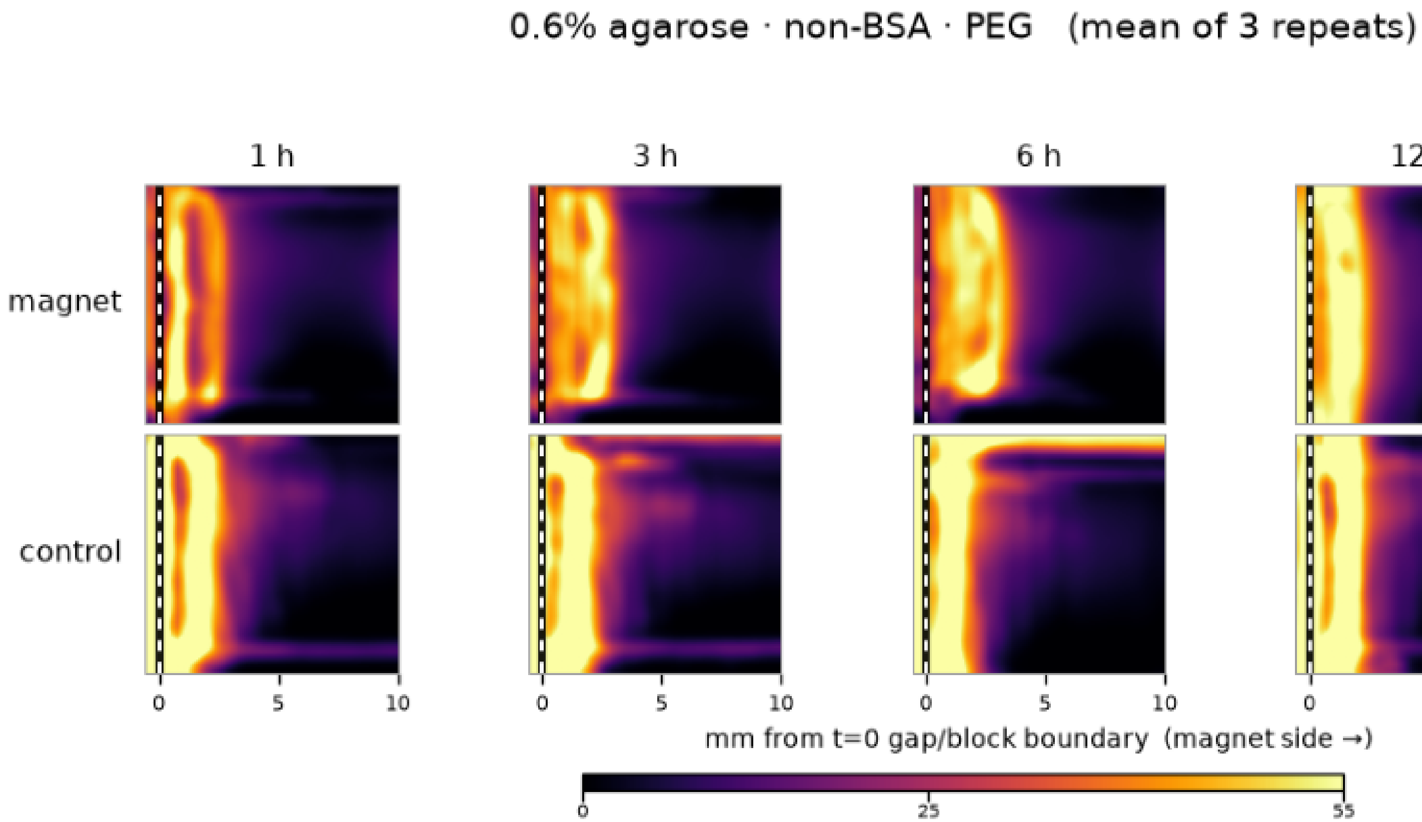

0.4% agarose · non-BSA · PEG (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55


0.6% agarose · BSA · COOH (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55



0.6% agarose · BSA · PEG (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
n=2
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55


0.6% agarose · non-BSA · COOH (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55


0.6% agarose · non-BSA · PEG (mean of 3 repeats)
1 h
3 h
6 h
12 h
21.5 h
magnet
control
0
5
10
mm from t=0 gap/block boundary (magnet side →)
0
25
55